OXFORD
INTERNATIONAL
AQA EXAMINATIONS

INTERNATIONAL GCSE

BIOLOGY

9201/1

Paper 1

Mark scheme

November 2019

Version: 1.0 Final

19BY92011/MS

Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Assessment Writer.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Further copies of this mark scheme are available from oxfordaqaexams.org.uk

Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student’s answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student’s answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

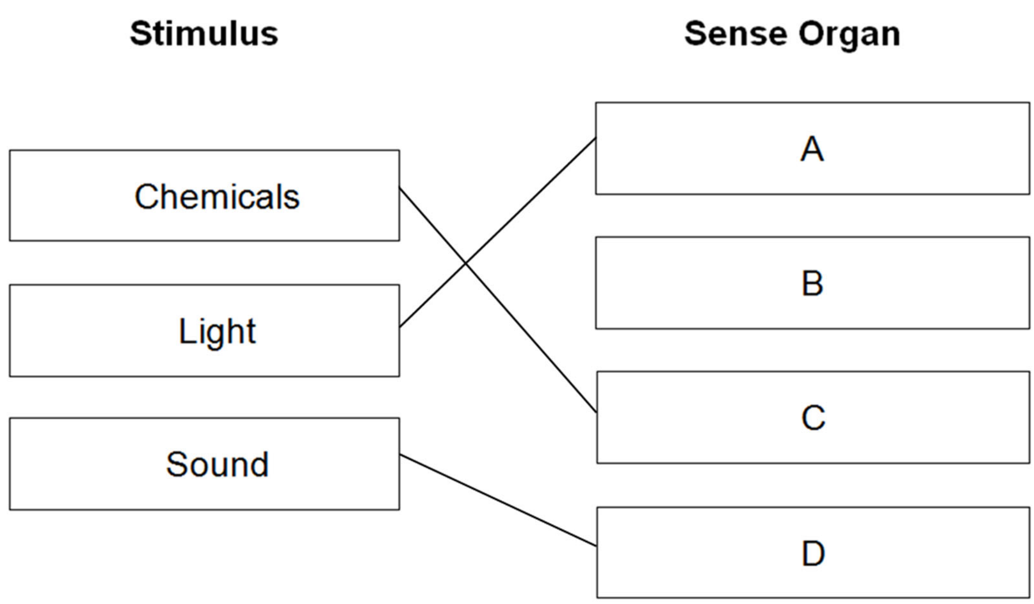
Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student’s answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner’s mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1			3	AO1 3.4.2c
01.2	automatic rapid	allow in either order	1 1	AO1 3.4.1c
01.3	pain-withdrawal reflex	allow description allow other correct examples	1	AO1 3.4.1d
01.4	sensory neurone		1	AO1 3.4.1d
01.5	any **one** from: • muscle(s) • gland(s)	allow named effector	1	AO1 3.4.1e
01.6	a chemical	ignore impulse	1	AO1 3.4.1d
01.7	brain spinal cord	in either order	1	AO1 3.4.1d

Total			**10**

Question	Answers	Extra information	Mark	AO/Spec. Ref.
02.1	(A) nucleus (B) cytoplasm		1 1	AO2 3.1.1a
02.2	$\frac{25}{2000}$ 0.0125 (mm)	an answer of 0.0125 (mm) scores **2** marks. allow 0.013	1 1	AO2 3.1.1a 6.3.10
02.3	any **two** from: • bacteria • virus • protist		2	AO1 3.4.7b/c
02.4	any **one** from: • mumps • rubella / german measles	allow phonetic spelling ignore measles	1	AO1 3.4.7e
02.5	lower percentage of measles (if two doses)	allow higher percentage of measles if only one dose received	1	AO3 3.4.7e
02.6	fewer cases (of measles if given at 12-18 months) for both 1 dose **and** 2 doses		1 1	AO3 3.4.7e
Total			**10**	

Question	Answers	Extra information	Mark	AO/Spec. Ref.
03.1	any **two** from: • food • mates • territory		2	AO1 3.3.2c
03.2	any **two** from: • light • water (from soil) • nutrients (from soil)	ignore sun allow moisture / rain allow minerals / ions / named ion	2	AO1 3.3.2b
03.3	(for crop plants) less light **or** less water (so) less photosynthesis (therefore) less carbohydrate / glucose formed **OR** less nitrate (ions) for crop plants (1) (so) less protein formed (1) (so) less growth (1)	ignore food allow sugar for glucose allow other correct organic molecules if no other mark awarded allow less minerals / mineral ions for **1** mark	1 1 1	AO1 AO2 3.3.2b 3.2.1c 3.2.1d
03.4	enzyme		1	AO1 3.5.5b
03.5	any **two** from: • gene inserted into a vector • vector used to insert gene into cells (of organism B) • at an early stage of development / embryo	allow plasmid or virus for vector	2	AO1 3.5.5b

03.6	(the herbicide) kills the other plants / weeds but not the crop (so) less competition (and increased yield of crop)	allow kills <u>only</u> other plants / weeds allow the crop has access to more light / water / minerals / ions	1 1	AO2 3.5.5e/d

03.7	(possible) effect on populations of wild flowers / insects (possible) effects on human health (if eaten)	allow examples/descriptions allow lack of genetic variation could lead to disease wiping out crops and causing famine. do **not** accept reference to harmful chemicals	1 1	AO1 3.5.5f

Total			**14**

Question	Answers	Extra information	Mark	AO/Spec. Ref.
04.1	muscle elastic	allow in either order	1 1	AO1 3.2.3j
04.2	blood at high**er** pressure	allow converse for veins	1	AO2 3.2.3j
04.3	thinner / very thin (walls)	allow no muscle / elastic tissue allow (only) single layer of cells	1	AO1 3.2.3k
04.4	any **three** from: • (abnormal) haemoglobin may not combine as easily with oxygen (in the lungs) • (abnormal) haemoglobin may not release oxygen as easily (in the tissues) • (red) blood cells may get stuck in small blood vessels / capillaries **or** may clump / stick together in capillaries causing blockages / cell death • (so red) blood cells less likely to reach lungs / alveoli to pick up oxygen • (and so red) blood cells less likely to reach tissues to release oxygen • reduced surface area for oxygen to combine with haemoglobin	allow spleen removes (red) blood cells so haemoglobin level drops	3	AO3 3.2.3n

04.5	correct parental genotypes **or** gametes: **Aa** <u>and</u> **Aa** **or** **A** and **a** <u>and</u> **A** and **a**	allow alternative symbols **only** if defined	1	AO2 3.5.3g 3.5.4a 6.3.6
	offspring genotypes correctly derived**:** **AA Aa Aa aa**	allow (offspring) genotypes correct for student's parental genotypes / gametes	1	
	identification of **aa** as having sickle cell disorder		1	
	correct probability: 0.25 / $^{1}/_{4}$ / 1 in 4 / 25% / 1:3	do **not** accept 3:1 / 1:4 for probability	1	

04.6		an answer of 0.36 scores **3** marks.		AO2 3.2.3j
	$\pi \times 0.34^2$		1	
	0.362984	allow 0.3631681	1	
		allow correct calculation from incorrect formula	1	
	0.36 (cm^2)	allow 1 mark for an incorrectly calculated value given to 2 significant figures		

04.7		an answer of 0.083 scores 3 marks.		AO2 3.2.3j
	0.012 x 2.0 = 0.024 (cm^3)		1	
	$\frac{0.024}{0.29}$		1	
	0.083 (cm^3 /s)	allow 0.08 / 0.0828	1	

04.8	any **four** from: • arteries become blocked **or** blood clot forms in arteries • high blood pressure causes arteries to rupture • bleeding in the brain / stroke • stops / not enough blood flow (to brain cells) • not enough oxygen / glucose (to brain cells) • no/less respiration **or** no/less energy released		4	AO2 3.2.6a/c 3.2.3a/i/n

Total			**21**

Question	Answers	Extra information	Mark	AO/Spec. Ref.
05.1	trachea	allow windpipe ignore bronchus	1	AO2 3.2.5a
05.2	contracting increase lower / less than inflate / enter / move into		1 1 1 1	AO2 3.2.5b
05.3	any **two** from: • cells do not spread in the blood • do not invade other tissues • do not form secondary tumours	allow converse for malignant tumours spread alone insufficient	2	AO1 3.5.2n
05.4	pipette higher resolution	allow smallest scale divisions ignore more accurate	1 1	AO4 3.2.3
05.5	as a control / comparison	allow to set a base line	1	AO4 3.2.5
05.6	the reaction had stopped **or** there was no more CK-19 to bind		1	AO4 3.2.3
05.7	only the CK-19 protein is the (right) shape to match / fit that of the antibody		1	AO2 3.4.7d 3.2.3r

05.8	$\left(\frac{(4.1\ x\ 10^{-6})+(6.3\ x\ 10^{-6})+(6.9\ x\ 10^{-6})}{3}\right)$	an answer of 4.1×10^{-6} scores **3** marks		AO2 3.5.2.n 6.3.7 6.3.15
	$= 5.8 \times 10^{-6}$		1	
	$(5.8 \times 10^{-6}) \times 0.7$		1	
	4.1×10^{-6} (mg)	allow 0.0000041	1	

05.9	**Level 3:** A judgement, strongly linked and logically supported by a sufficient range of correct reasons, is given.	5-6	AO3 3.5.2n
	Level 2: Some logically linked reasons are given. There may also be a simple judgement.	3-4	AO3 3.5.2n
	Level 1: Relevant points are made. They are not logically linked.	1-2	AO3 3.5.2.n
	No relevant content	0	
	Indicative content allow ref to candidates answer to question 05.8. **For:** • mean level found in patients with tumours is higher by 3.6×10^{-6} mg. • the highest mean levels of CK-19 are only found in tumour patients. • All groups with tumours had a higher level of CK-19 than those groups with no tumours. **Against:** • people without tumours still have CK-19 in their blood. • these are mean values and the highest mean of 2.9×10^{-6} mg in non-tumour patients is only 1.2×10^{-6} mg less than mean for tumour patients in group 1. • CK-19 could be due to other forms of lung damage. • CK-19 may also be produced by other tumours • CK-19 may also be produced by normal cells • does not refer to lungs, or tumours in the lungs, only to CK-19 as a tumour marker • small number of people in study • values for mean may cover wide ranges which may overlap		

Total			**21**

Question	Answers	Extra information	Mark	AO/Spec. Ref.
06.1	keeping conditions (in the body) within narrow limits / constant		1	AO1 3.4.2a
06.2	blood is filtered		1	AO1 3.4.3d
	all the glucose is reabsorbed		1	
	some dissolved ions are reabsorbed		1	
	some / most of the water is reabsorbed		1	
	urea, excess / some ions and excess / some water remain to form urine		1	
06.3	70 without CHF **and** 20 with CHF		1	AO2 AO3 3.2.3b
	(so) reduced stroke volume but no increase in heart rate (so reduced cardiac output).		1	
6.4	pituitary gland		1	AO1 3.4.3e

06.5	increased levels of ADH (so) more water is reabsorbed by the kidneys. decreased blood flow through the kidneys (so) less water filtered out of the blood. (so) more water retained in the blood increasing blood volume (and) greater volume will result in increased blood pressure due to increased blood volume, muscles of ventricles will be unable to contract properly		1 1 1 1 1	AO2 AO3 3.4.3e 3.2.3 b/c/h

Total			**14**